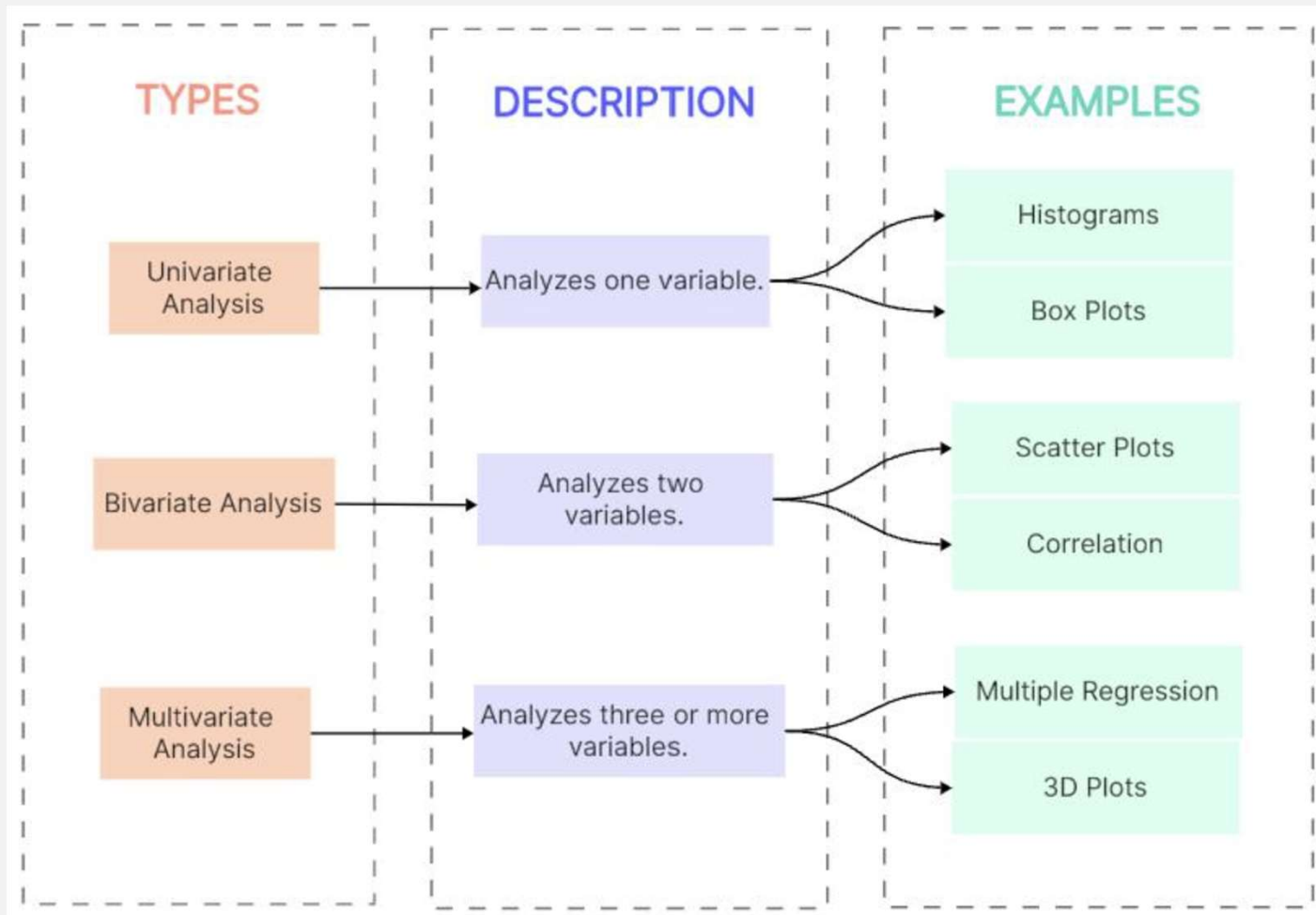


DATA ANALYTICS

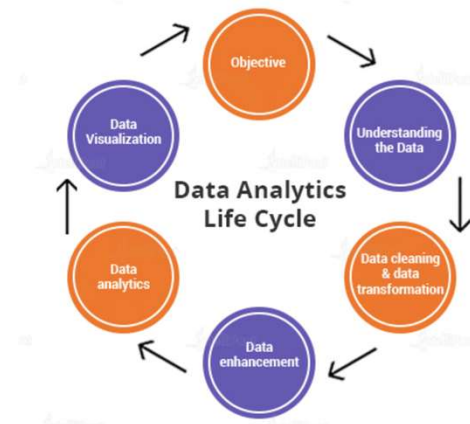
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Difference B/w Univariate, Bivariate & Multivariate Analysis

<u>Basis for Diff</u>	Univariate	Bivariate	Multivariate
Focus	Univariate analysis examines a single variable at a time	Bivariate analysis examines the relationship between two variables	Multivariate analysis involves the simultaneous analysis of three or more variables.
Objective	The objective is to describe and understand the characteristics, distribution, and variability of the variable.	The objective is to determine if there is a relationship, association, or correlation between the two variables.	The objective is to understand complex relationships among multiple variables, considering interactions and dependencies between them
Examples	Descriptive statistics such as mean, median, mode, variance, standard deviation; graphical representations like histograms, bar charts, and box plots.	Scatter plots, correlation analysis (e.g., Pearson correlation coefficient), chi-square tests, crosstabulations, simple linear regression.	Multivariate regression analysis, principal component analysis (PCA), factor analysis, cluster analysis, multivariate analysis of variance (MANOVA), canonical correlation analysis (CCA).
Application	Commonly used for preliminary exploration of data and understanding the properties of individual variables.	Used to explore the connection between two variables and understand how changes in one variable are related to changes in another.	Used to uncover patterns, identify underlying structures, and analyze complex relationships among multiple variables in data

STEPS OF DATA ANALYTICS



6 Steps of Data Analysis



STEPS OF DATA PROCESSING

Data Collection

- Accuracy
- Completeness
- Real-time Data

Data Preprocessing

- Cleansing
- Transformation
- Deduplication
- Normalization

Data Storage

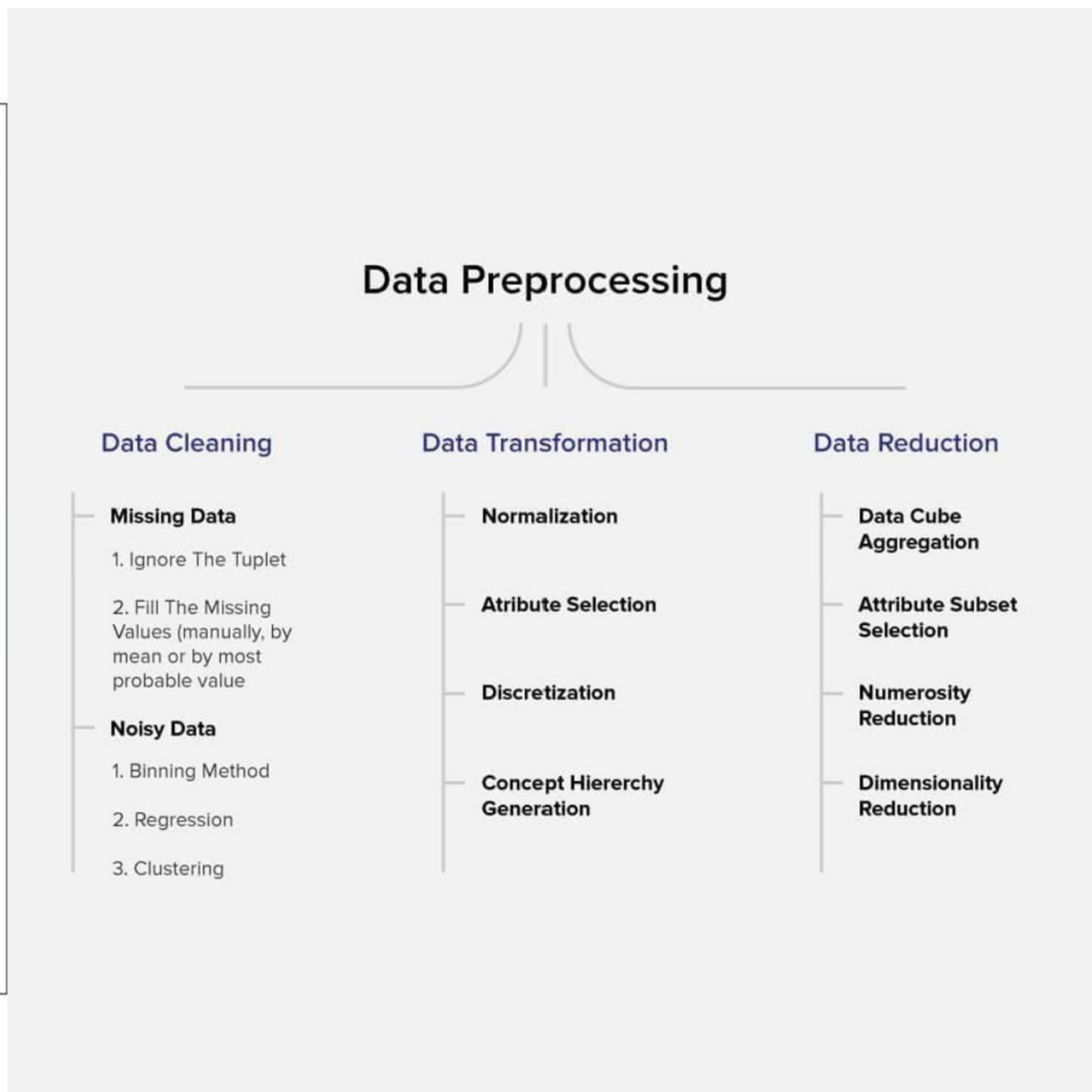
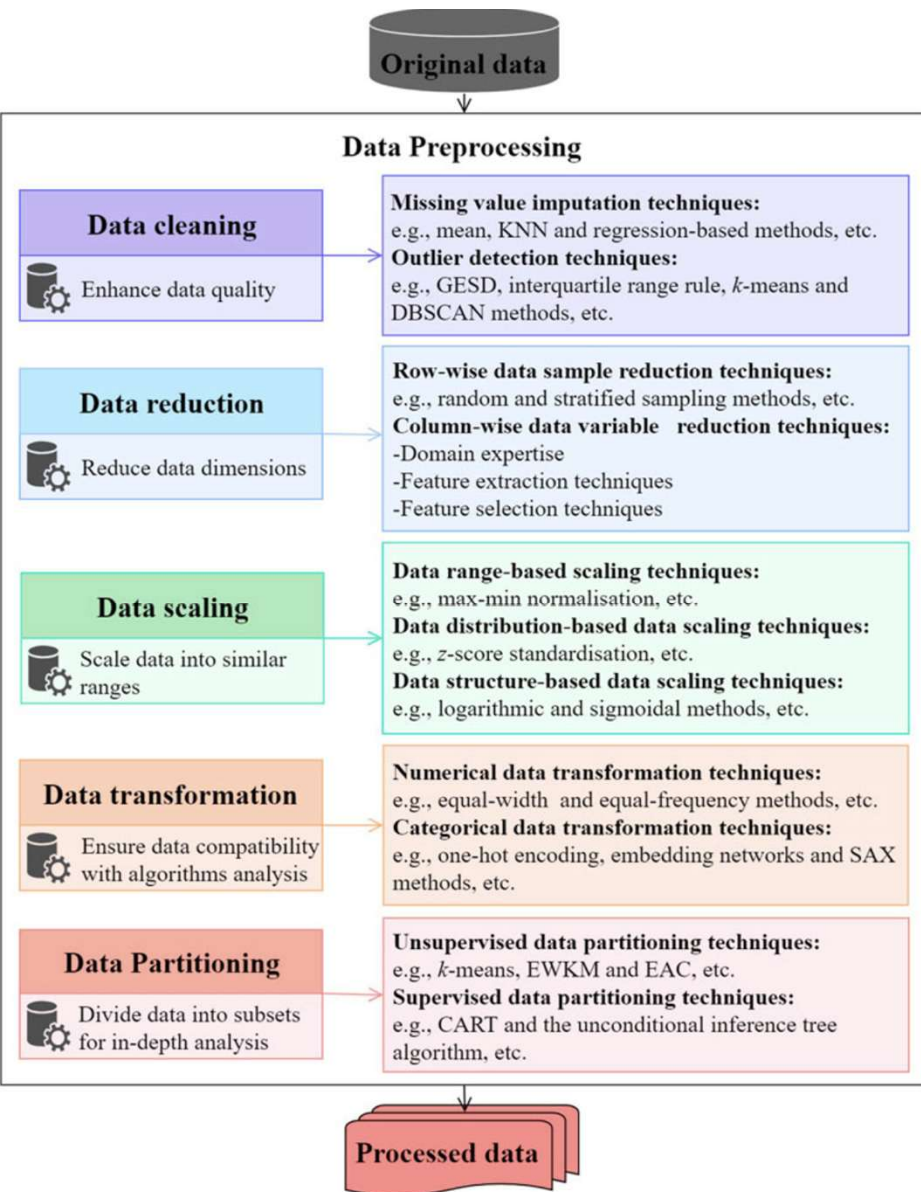
- Scalability
- Redundancy
- Data Retrieval

Data Analysis

- Descriptive Analytics
- Predictive Analytics
- Prescriptive Analytics

Data Visualization

- Clarity
- Interactivity
- Relevance



REASONS FOR MISSING DATA

- Errors during data collection or entry
- Incomplete recording of information
- Respondents not providing certain data

MISSING DATA

- Missing values can be broadly classified into the following 3 types:
 1. **Missing Completely at Random (MCAR)** — this type of missing value does not have any relation with any other value in the dataset. There is no pattern or dependency on other values. This could be due to some human error, error in data loading, failure of system, etc.
 2. **Missing at Random (MAR)** — This is the type of missing value that has some sort of relationship or dependency on other values in the dataset but not with the missing data. A pattern can be observed in the missing data and hence, the probability of missing values is dependent on other values. For example, a dataset having a column 'drinks alcohol' might have missing values for people having less age, as people below the age of 21 might not be comfortable sharing this information.
 3. **Missing Not at Random (MNAR)** — The type of data is missing not at random (MNAR) if the pattern in the missing values is not dependent on the other values in the dataset. For example, it might be because some people are unwilling to answer some questions in a survey. Like, people who are overweight might not be willing to share their weight.

DATA IMPUTATION

- Data imputation is the process of replacing missing values in a dataset with estimated values.
- a statistical technique used to handle incomplete datasets
- allows for more complete analysis and prevents the loss of data points that would occur if rows with missing values were simply deleted.
- deleting rows with missing data can introduce bias and reduce the efficiency of the analysis, so imputation is preferred as it preserves all cases.

Missing Data Mechanisms

- MCAR – $p(\text{missing})$ is *unrelated* to all variables, observed and unobserved

$$p(\text{missing}|\text{complete data}) = p(\text{missing})$$

- MAR – $p(\text{missing})$ is related to observed variables [observed data] only

$$p(\text{missing}|\text{complete data}) = p(\text{missing}|\text{observed data})$$

- MNAR – $p(\text{missing})$ is related to the unobserved/missing variables [missing data]

$$p(\text{missing}|\text{complete data}) \neq p(\text{missing}|\text{observed data})$$

(see Schafer & Graham, 2002)²³

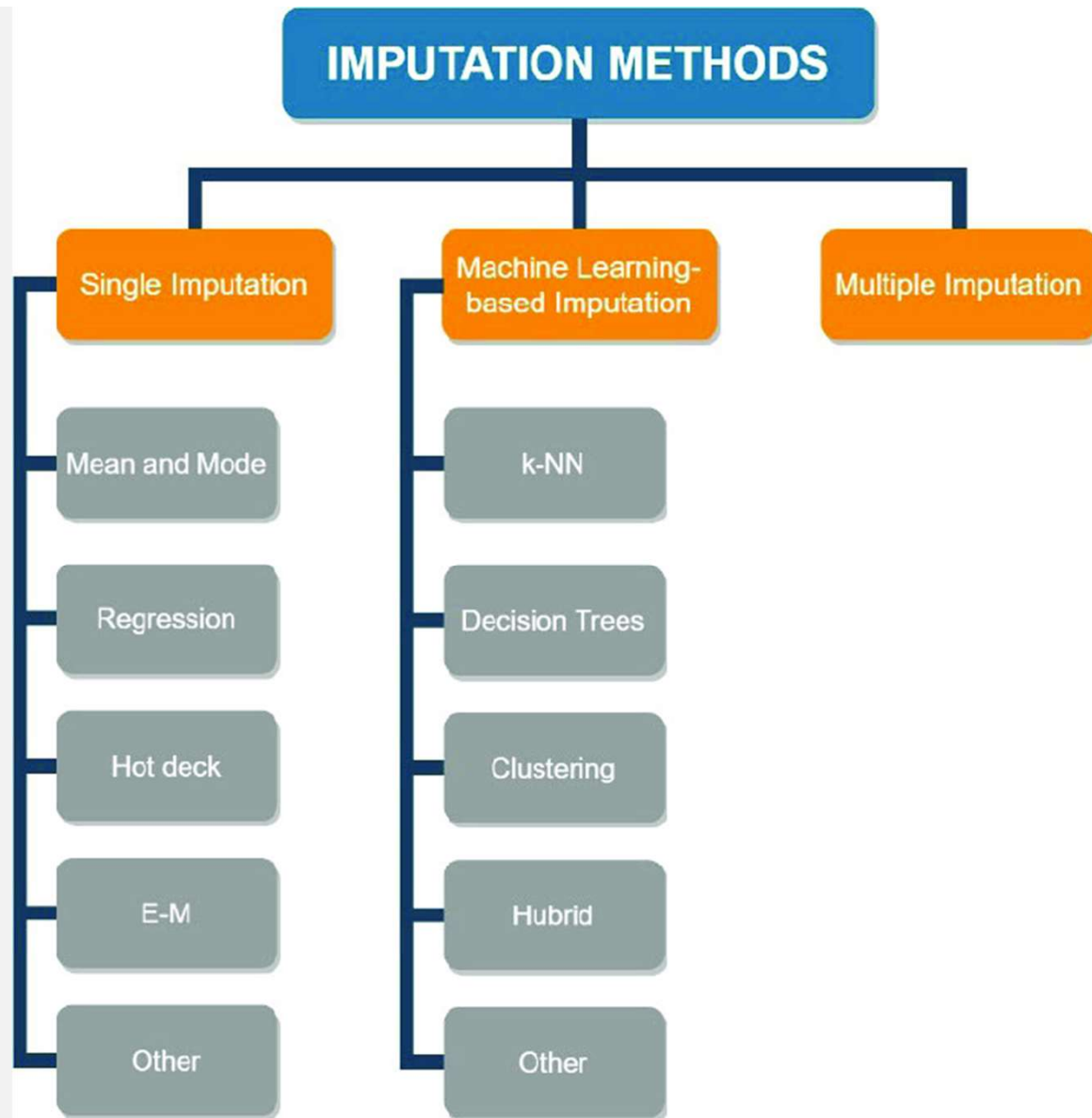
Aspect	Missing at Random (MAR)	Missing Not at Random (MNAR)
Dependency	The probability of missingness depends on observed data	The probability of missingness depends on unobserved data
Example	A survey where the likelihood of skipping a question about income depends on the respondent's age (observed)	A survey where the likelihood of skipping a question about income depends on the respondent's actual income (unobserved)
Implication	Analysis can be adjusted using observed data	Analysis requires models that account for the reason behind missingness, which is more complex

HOW TO DEAL WITH MISSING DATA?

1. Deletion (rows, columns)

2. Imputation

- Mean, median, mode
- Machine learning based: Linear Regression, KNN, EM, ...



MULTIVARIATE IMPUTATION OF CHAINED EQUATIONS (MICE)

- Stef van Buuren introduced Multiple Imputation by Chained Equations (MICE) to overcome the limitation of missing data in **univariate** perspectives.
- Impute values through iterative estimations based on conditional distributions.
- Account for dependencies between variables, by using other variables to estimate the missing data.

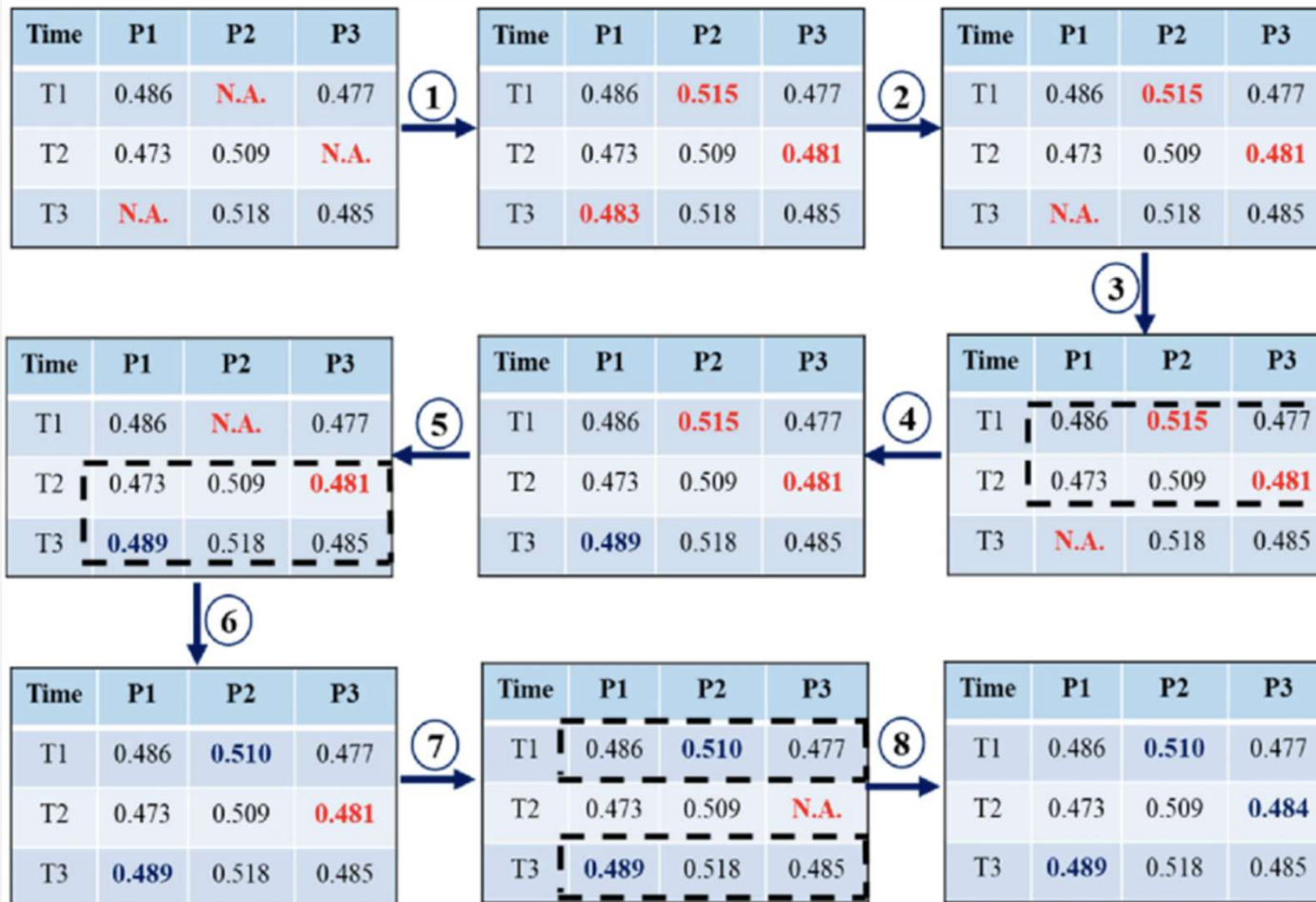
MULTIVARIATE IMPUTATION OF CHAINED EQUATIONS (**MICE**)

- Stef van Buuren introduced Multiple Imputation by Chained Equations (MICE) to overcome the limitation of missing data in **univariate** perspectives.
- Impute values through **iterative** estimations based on conditional distributions.
- Account for **dependencies** between variables, by using other variables to estimate the missing data.

MICE ALGORITHM

- How MICE operates step-by-step:
 - 1.Initial Imputation:** Fill in all the missing values in the dataset with simple placeholders, for example the mean or median of each variable (for categorical type, the most frequent will do).
 - 2.Reset Missing Values:** Next, **choose one variable** at a time for imputation, and change those placeholder values back to missing.
 - 3.Build Regression Model:** Build a regression model to predict the variable from step 2, by using the other independent variables in the dataset. This model uses all (or some) of the other variables in the dataset as predictors to estimate the values of the missing data points.
 - 4.Replace Missing Values:** Use the regression model to predict and fill in the missing values for this variable. At this point, this variable has now both observed and newly imputed values.
 - 5.Repeat for Each Variable:** Repeat steps 2-4 for every variable in the dataset that has missing values.
 - 6.Repeat steps 2-5 for several cycles.**
 - Going through all the variables will complete one iteration or cycle
 - In general, after 10 cycles, the model is said to have reached convergence and the estimates don't improve any further.

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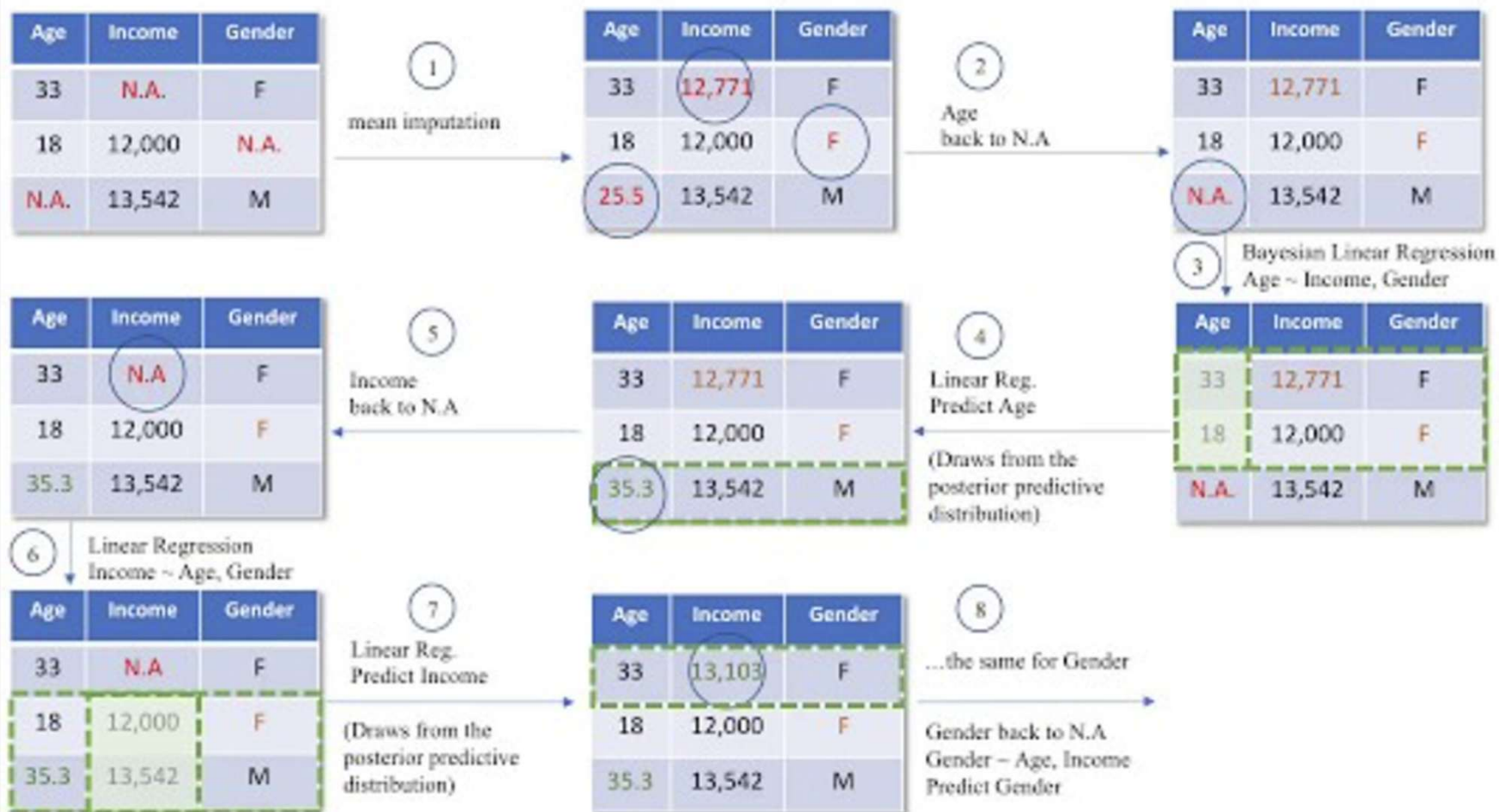


P1, P2, P3 – Pressure Values

T1, T2, T3 – Instantaneous time intervals

N.A. – Not Applicable (Represents missing values)

Multiple Imputation by Chained Equations (MICE) – Single Iteration



Ofir Shalev (@ofird) May 2018

Age	Income	Gender
33	NA	F
18	12000	NA
NA	13542	M



Age	Income	Gender
33	12771	F
18	12000	F
NA	13542	M

Take mean, keep missing value in Age



Age	Income	Gender
33	12771	F
18	12000	F
35.3	13542	M

Predict Age ~ Income, Gender



Age	Income	Gender
33	13103	F
18	12000	NA
35.3	13542	M

Reinstate missing values in Gender



Age	Income	Gender
33	13103	F
18	12000	M
35.3	13542	M

Predict Gender ~ Age, Income



Age	Income	Gender
33	13103	F
18	12000	F
35.3	13542	M

Predict Income ~ Age, Gender



Age	Income	Gender
33	NA	F
18	12000	F
35.3	13542	M

Reinstate missing values in Income



Age	Income	Gender
33	13103	F
18	12000	M
NA	13542	M

Reinstate missing values in Age



Repeat for n times

FEATURE SELECTION

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FEATURE SELECTION

- Feature selection is important because it can:
 - Shorten training times
 - Simplify models and make them easier to interpret
 - Enhances testing set performance by reducing overfitting

DROPPING A FEATURE

- Features that can be dropped without impacting the model's precision significantly fall into **2** categories:
- **Redundant**: This is a feature that is highly correlated to other input features and therefore does not add much new information to the signal.
- **Irrelevant**: This is a feature that has a low correlation with the target feature and for that reason provides more noise than signal.

WRAPPER-BASED METHODS

- Feature selection is essentially reduced to a search problem using these steps:
 - 1. A subset of features is used to train a model
 - 2. Based on the results of each iteration, features are added to or removed from the subset

Forward Selection:

1. Starts by having no features in the dataset. (**empty** set)
2. During each iteration, features are added with the intent of improving the performance of the model.
3. If performance is **improved**, the features are **kept**.
4. Features that do not improve the results are discarded.
5. The process continues until improvement of the model stalls.

Backward Elimination:

1. All features are initially present in the dataset.
2. Features with the least significance are removed during each iteration
3. The process then checks if the performance of the model improves.
4. The process is repeated until no significant improvement is observed.

Recursive Feature Elimination (RFE):

- **Algorithm:**

1. trains a model (using linear regression, SVM, or random forest, ...) with **all** features,
2. **ranks** them by importance (e.g., using `feature_importances_` in tree-based models),
3. removes the least important feature(s).
4. This is repeated until the desired number of features is reached.

- **Removal criteria:** Based on a model's **feature importance** metric, like coefficients, impurity decrease, or other specific importance measures.

Recursive Feature Elimination (RFE):

- RFE is best when a model provides a clear and reliable feature importance score, such as **linear** models or **tree-based** models

RFE (Recursive Feature Elimination)	Backward Elimination
Removes features based on their importance score from the model.	Removes features based on the impact on model performance metrics.
Iterative model training and feature importance ranking.	Iterative model training and performance evaluation after removing each feature.
Model-derived feature importance (e.g., coefficients, impurity).	Performance metric change (e.g., R-squared, AIC).

FEATURE IMPORTANCE

PICK A FEATURE BASED ON ITS
IMPORTANCE

RECURSIVE FEATURE ELIMINATION (RFE)

How Recursive Feature Elimination Works



Step-by-Step Process

- 1. Start with All Features:** RFE begins with the complete set of features in your dataset.
- 2. Train a Model:**
 1. A specified estimator (e.g., a **linear regression** model, **decision tree**, or support vector machine) is trained on the entire feature set.
 2. The estimator must be able to rank features by their importance (e.g., weights, coefficients, or other metrics).
- 3. Rank Features by Importance:** After training, the model assigns an importance score to each feature. For instance:
 1. In a **linear regression**, **coefficients** indicate **feature importance**.
 2. In a **decision tree**, **feature importance** is derived from **split criteria**.
- 4. Remove the Least Important Feature(s):** The feature(s) with the lowest importance score are removed from the dataset.
- 5. Repeat the Process:** The model is re-trained on the reduced feature set, and the elimination process is repeated until the desired number of features remains.
- 6. Finalize the Selected Features:** At the end of the process, RFE outputs the optimal subset of features, ranked by their importance.